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Inventor(s)

Benedict; Michael et al.

Handheld filament extension atomizer for precision delivery of drugs and therapeutics

Abstract

A docking station for a hand-held filament extension atomizer device includes a receiver to receive the device, station electronics, a recharging point for the device arranged to connect with a power source of the device, a power connection to an alternating current power source, and a cleaning reservoir of cleaning solution. A method of operating hand-held dispenser to dispense fluid as a mist includes receiving a signal at a motor contained in a casing in response to a user triggering an actuator on the casing, activating the motor to provide fluid to a filament extension atomizer contained in the case from a reservoir contained in the casing, the motor to cause the filament extension atomizer to generate a mist, and using an air source contained in the casing arranged adjacent the filament extension atomizer to provide air flow to direct the mist to a nozzle that is arranged to allow the mist to exit the casing.

Inventors: Benedict; Michael (Palo Alto, CA), Johnson; David Mathew (San Francisco, CA), Solberg; Scott E. (San Jose, CA), Kalb; Jamie (Mountain View, CA), Neelakantan; Ravi (Redwood City, CA), Unidad; Jerome (Paris, FR), Taylor; David Charles (San Francisco, CA), Yan; Frances (San Francisco, CA)

Applicant: PALO ALTO RESEARCH CENTER INCORPORATED (Palo Alto, CA)

Family ID: 68066700

Assignee: XEROX CORPORATION (Norwalk, CT)

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Primary Examiner: Weng; Kai H

Attorney, Agent or Firm: Miller Nash LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation of, and claims priority to, U.S. patent application Ser. No. 16/140,902, filed Sep. 25, 2018, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) This disclosure relates to atomization of fluids, more particularly to hand-held atomizers for drug and therapeutics delivery.

BACKGROUND

(2) The primary method for the delivery of eye drops has been an eye drop dispenser, also referred to as a droptainer. These small containers typically have an orifice with a controlled size that regulates how much liquid comes out when the container tips upside down.

(3) However, many users find eye drops difficult to use and would welcome alternative methods to deliver materials to the eye. Additionally, since a droptainer delivers only a single drop as one large droplet, much of the volume of material delivered is lost. The delivered volume may only have 10% of the volume of the active material from the droptainer.

(4) Spray delivery provides a method for the delivery of these drugs. Spray delivery can overcome many of the challenges associated with a droptainer since additional momentum imparted to the spray particles allows the delivery device to work at any angle relative to the eye. However, existing spray delivery systems have their own challenges. One approach, pneumatic atomization, may result in large globs of spray during the beginning of the stroke. Additionally, when non-Newtonian, extensionally hardening fluid is used a pneumatic actuator will produce a filament like stream of fluid, not a mist of small droplets. Ultrasonic and vibrating mesh technologies can produce a fine, small mist without large droplets but have extreme limitations on rheology and cannot process fluids at all that have even small amounts of extensionally hardening properties.

SUMMARY

(5) According to aspects illustrated here, there is provided a docking station for a hand-held filament extension atomizer device includes a receiver to receive the device, station electronics, a recharging point for the device arranged to connect with a power source of the device, a power connection to an alternating current power source, and a cleaning reservoir of cleaning solution.

(6) According to aspects illustrated here, there is provided a method of operating a hand-held dispenser to dispense fluid as a mist includes receiving a signal at a motor contained in a casing in response to a user triggering an actuator on the casing, activating the motor to provide fluid to a filament extension atomizer contained in the case from a reservoir contained in the casing, the

motor to cause the filament extension atomizer to generate a mist, and using an air source contained in the casing arranged adjacent the filament extension atomizer to provide air flow to direct the mist to a nozzle that is arranged to allow the mist to exit the casing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 shows an embodiment of hand-held fluid dispenser.
- (2) FIG. 2 shows an embodiment of a replaceable cartridge for a hand-held fluid dispenser.
- (3) FIGS. 3 and 4 show embodiments of a hand-held fluid dispenser with a replaceable fluid cartridge.
- (4) FIGS. 5 and 6 show an embodiment of a replaceable head cartridge for a hand-held fluid dispenser.
- (5) FIGS. 7 and 8 show an alternative embodiment of a replaceable head cartridge for a hand-held dispenser.
- (6) FIGS. 9-11 show various embodiments of drive shafts for a replaceable head cartridge.
- (7) FIG. 12 shows an embodiment of a replaceable head cartridge having connection tubes.
- (8) FIG. 13 shows an embodiment of a filament extension atomizer roller having a hub motor.
- (9) FIG. 14 shows an embodiment of an impeller attachable to a motor in a hand-held fluid dispenser.
- (10) FIGS. 15 and 16 show an embodiment of a casing and internal rollers for a hand-held fluid dispenser.
- (11) FIGS. 17 and 18 show an embodiment of a casing, internal rollers, and an impeller for a hand-held fluid dispenser.
- (12) FIG. 19 shows an embodiment of a motor mount and a mountable impeller for a hand-held fluid dispenser.
- (13) FIG. 20 shows an embodiment of a drive shaft having shared but separated roller and air generating components.
- (14) FIG. 21 shows a flowchart of an embodiment of a cleaning process for a dispensing device.
- (15) FIGS. 22-27 show embodiments of a docking and cleaning station for a hand-help fluid dispenser.

DETAILED DESCRIPTION OF THE EMBODIMENTS

- (16) FIG. 1 shows an embodiment of a hand-held dispenser to dispense fluid as a mist using a filament extension atomizer. The hand-held dispenser has a casing 12 configured to fit in a user's hand. The casing contains all the elements of the dispenser including the nozzle 14, the liquid reservoir 16, a filament extension atomizer 18, an air source 20, a motor/drive unit 22, an actuator 24, control electronics and circuitry 26, a power source 28, and an optional flow control 30. The casing has dimensions and a form factor designed to fit into a user's hand.
- (17) The filament extension atomizer 18 receives a fluid from a fluid reservoir 16 and delivers a mist to the nozzle 14. The nozzle 14 may have a dimension selected to focus the spray on a target location of a specific size. For best results, the area is tightly controlled and as small as possible to minimize overspray, maximize delivery, and prevent undesirable contact with the skin. The fluid to be delivered may consist of some sort of therapeutic material, such as eye drops, antibiotic sprays for wounds, etc.
- (18) The fluid reservoir 16 contains the fluid. The reservoir may be refillable or replaceable, as will be discussed in more detail later. In addition, more than one reservoir may be present in the casing, with a selector knob or other means of choosing which fluid reservoir sends fluid to the filament extension atomizer. The reservoir may be pressurized, such as with air, or mechanical compression such as a spring. The liquid reservoir may interface with the filament extension atomizer and the

other components of the system through a port, tube, valve, etc.

(19) The filament extension atomizer generates a mist from the fluid under control of an electronic control circuit and powered by the motor/drive circuit **22**. The air source **20** directs the mist from the filament extension atomizer to the nozzle. Since small particles (under 100 microns) are produced by the Filament Extension Atomizer, they will quickly lose momentum once they have exited the device. The air source could be an electronic air pump, a fan, a compressed container, or any other source of air volume. The air helps direct the spray towards the surface being treated and helps maximize delivery efficiency. The air speed and pressure should be minimized to maximize comfort. The choice of airflow speed and pressure may depend on the application area. For example, a close-range application, such as a range of less than 25 millimeters, requires low airspeeds to allow the maximum amount of material to reach the substrate in a small area. However, if the application uses a longer distance, the longer distance to the target will require more momentum and therefore higher air speeds. Air is activated through the use of a pump or a valve. A pump may be used if no stored air is used, in this case the pumping element is activated either through an electronic control or through the motion of the driving element to create pressure from the air. If an onboard air source is used, such as a compressed gas, the valve is actuated either through an electronic signal or the motion of the actuation drive unit to release the compressed air in a controlled manner.

(20) The filament extension atomizer **18** converts the fluid to a mist by stretching the fluid between diverging surfaces to form filaments. The diverging surfaces cause the filaments to break up into droplets that form the mist. The filament extension atomizer may use many different types of diverging surfaces. The embodiments here employ two counter rotating rollers. As the surfaces of the rollers rotate away from each other, the fluid forms filaments that then extend to the point of breaking into a mist.

(21) The filament extension atomizer runs under control of the motor **22**. This may consist of an electric motor designed to operate at high speed, typically thousands to tens of thousands of revolutions per minute. The motor will couple to the filament extension atomizer through mechanical couplings. This could include belts, pulleys, gears, a shaft, or electrical or magnetic coupling. The filament extension atomizer may also include a gearing mechanism to drive the rollers or motors at different speeds.

(22) The device operates when a user presses or otherwise activates the actuator **24**. This may consist of a button or other actuator that causes a signal to be sent to the control electronics. For user convenience, the actuator **24** may reside in a position on the casing **12** such that the user can hold the dispenser and activate the actuator with the same hand. The button may have multiple positions or multiple sensing modes. For example, a button may initially be depressed slightly or detect contact through capacitive means to activate the device's electronics or turn on some of the subsystems and then when the device is further depressed the system may dispense a dose. Alternatively, multiple sensing modes, such as capacitive and physical depressing can be used to allow two activation modes to be used.

(23) Upon receiving a signal from the actuator, the electronic control circuitry causes the device to operate. The control circuit may control the charging of the battery or other power source **28**, provide drive voltages to the motor, switch any pumps or valves, and provide user feedback and control. The electronic control circuitry may consist of a controller integrated circuit, a circuit board with the necessary components to manage the control, a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), a microcontroller, etc. The actuator may actuate components either simultaneously or in a specific order. In some embodiments, the air is turned on first, followed by a short delay, the motor is activated to turn on the rollers, after another short delay the pump is activated for a fixed period of time. When the pump has been deactivated, the motor is turned off, followed by a short delay, followed by the airflow.

(24) The electronics may include a communications link to allow the dispenser to communicate

with device external to the dispenser through common wireless protocols such as Bluetooth, WiFi, or other near-field communications. The battery **28** will typically consist of a rechargeable battery and may be charged by a cord, contacts, or a wireless inductive charging system.

(25) FIG. **1** also shows a pump/flow control module **30** that may control the dosing of the liquid. This may include a pump or other means of flow control, such as a piezoelectric pump, under control of the electronic control circuitry. The pump may be optional, however, if the reservoir is pressurized. The dosing would be controlled by a valve instead of the pump. The air source previously described can also be utilized to provide a driving pressure to the reservoir with control being achieved by a valve.

(26) The dispenser shown in FIG. **1** allows the user to easily and accurately dispense a mist of fluid onto a target area. To make the dispenser more convenient for the user, as well as ensuring the proper operation of the device, some of the components may be replaceable. FIGS. **2-4** show an embodiment of a dispenser in which the fluid reservoir **16** consists of a replaceable fluid cartridge. As shown in FIG. **2**, the fluid cartridge may include a valve/seal **32**. In one embodiment, the seal may have a breakable inner surface that a structure or tube on the dispenser will break. The inner seal protects the contents of the cartridge from external contamination when the cartridge is not loaded. A seal ring then forms a liquid tight seal with the cartridge to avoid leakage and contamination. The seal structure may also include a valve that the control electronics can actuate. This may operate similarly to inkjet cartridges that have a small electrically-controlled valve that opens to allow ink to exit the inkjet cartridge and fall on paper or other print substrate.

(27) The fluid cartridge may contain multiple fluids simultaneously, separated by air or by a barrier of some kind, such as a film. As the device dispenses the fluid, it eventually exhausts all the therapeutic or drug fluid. The device will then reach the cleaning fluid and dispense cleaning fluid, cleaning the system. The user may be notified that is reaching the end of the useable fluid and cleaning fluid is about to be dispensed, by counting the number of doses dispensed from a cartridge or by detecting a change in the fluid, such as viscosity, optical opacity, or viscosity.

(28) Additionally, a fluid cartridge may contain multiple chambers separated by a divider. The divider may be a film or a solid wall. A cartridge with multiple chambers will include multiple seal and puncture structures and the device itself may have multiple valves or pumps to accommodate the multiple fluids. In this manner, the system may select from multiple fluids to dispense. For example, if a treatment involves multiple drugs, the system may dispense them sequentially to the user. Alternatively, one chamber may be a cleaning solution that the user can choose to activate if the device needs to be cleaned.

(29) The system may have multiple means for detecting information about the cartridges that have been loaded. A RFID or NFC tag can be placed on the cartridge in the form of a label or as a small component. The system electronics can be configured with electronics to read this data using the appropriate wireless protocols to detect information about the cartridge inserted. The information can include things like the material contained within the cartridge, the amount of fluid or doses, dose amount settings, settings for the FEA system, serial number, expiration date, or the recommended dose frequency for the user. The system can use this data to adjust settings of the airflow and the FEA spray system or provide information or prompts to the user to use the fluid at a certain frequency or time.

(30) FIGS. **3** and **4** show different options for the configuration of the dispenser. In FIG. **3**, the pump is used and has a drive shaft **34** that pumps the fluid from the reservoir to the filament extension atomizer. In FIG. **4**, typically with a pressurized container, only a conduit **36** is needed to transport the fluid from the reservoir to the filament extension atomizer **18**.

(31) In addition to replaceable fluid cartridges, other components may be replaceable, including the filament extension atomizer rollers. These removable cartridges, referred to here as head cartridges, contain the rollers. FIGS. **5-9** show embodiments of replaceable head cartridges. In FIG. **5**, the head cartridge system **40** contains the counter-rotating rollers **42** and **442** enclosed by head

cartridge casing **38**, referred to here as the head cartridge. This casing can be removed as a single unit and can easily be replaced. The head cartridges may have a communications link, such as NFC or RFID that allows the head cartridge to send information to the dispenser or one or more external devices, as discussed with regard to the cartridges above.

(32) FIG. **6** shows the replaceable head cartridge **38** in a horizontal arrangement that drops into the casing **12** to allow the rollers **42** and **44** to provide mist to the nozzle **14**. FIGS. **7** and **8** show a replaceable head cartridge with the rollers **42** and **44** in a vertical orientation. In this embodiment, the case **47** has an opening and a cover **45**. The head cartridges **38** slides into the opening and then the cover is closed. In this embodiment, a gear or spline **46** of FIG. **7** matches with a gear on the head cartridge to provide driving force for the rollers. FIG. **8** shows a similar vertical arrangement, but the drive force is supplied by shafts. FIGS. **9-11** show other options for drive connections.

(33) FIG. **9** shows a side view and a top view of a shaft **48** and its coupler **49**. FIG. **10** shows a d-shaped shaft **50** and its coupler **51**. FIG. **11** shows an example of a keys/keyway set up of the slot **54**, the key **53** and their coupler **52**. When the head cartridges connect to the dispenser, the connection needs to be a leak-resistant hydraulic connection. FIG. **12** shows a coupling tube **56** that snaps into the right reservoir **16** to allow transport of the fluid from the right reservoir **16**. The left reservoir may have a different coupling tube, not shown, to allow the user to select from different fluids.

(34) The rollers, in either the replaceable cartridge or not, have a similar size to the motor size. It is possible to utilize what is commonly referred to as a “hub motor” to combine the motor and roller subsystems. In this implementation the rotor is the roller and the stator is internal to the roller. The motor may consist of many different constructions, but a typical form would be a brushless DC motor with a permanent magnet (PM) rotor.

(35) FIG. **13** shows an embodiment of a hub motor **55**. The array of permanent magnets resides internal to the roller. The hub would have an electromagnetic stator, and there would be a bearing which puts the hub and roller/rotor in rotating mechanical communication. FIG. **13** shows a quarter section of this embodiment. The roller may have a coating or outer surface **58**. The permanent magnet array/rotor **60** resides inside the roller towards the outer edge, and the stator hub **62** resides inside the array of magnets. The outer surface **58** may be detachable from the rest of the assembly in an easy manner. This material may be replaced by the user or replaced by a professional service on a regular basis.

(36) The rollers rotate at speeds like many fans and blowers and have a similar diameter. Generation of a pressure differential directly on the surface of the rollers can induce air flow useful for the filament extension atomizer. One embodiment includes an impeller feature directly onto the radial face of the roller to create tangential air flow as shown with the impeller **64** of FIG. **14**.

(37) In this embodiment, the impeller pulls air in near the axis of rotation of the roller and blows it tangentially outwards in the radial direction. The filament extension atomizer may have additional features to direct the air both into the roller/blower and out of the tangential blades to create useful air flows. FIGS. **15** and **16** show external and internal views of the roller casing **38** containing the internal motors, combined roller blowers **42** and **44**, and air flow features on the filament extension atomizer system. The case **38** has air holes such as **66** to provide for air flow. FIGS. **17** and **18** show external and internal views of another embodiment of a casing **38** having external holes such as **66**, and the internal rollers such as **42**, one of which has an impeller **64**.

(38) FIG. **19** shows an impeller **64** mountable to the motor stator **68**, providing a system where the rollers and air generation system are easily replaceable. The roller and impeller can be removed and reattached the motor stator to facilitate cleaning or to be replaced. FIG. **20** shows an alternative arrangement where the motor **22** has a common shaft **34** with the roller **42**, and a separate impeller **64**. Alternatively, the impeller can be mounted on a separate shaft, connected to the motor through a coupling but not connected to the same drive shaft as the rollers. This may allow the impeller to operate at speeds other than the same speed as the roller, which will allow for airspeed and pressure

to be modulated separately.

(39) The use of a motor may allow the system to track the cleanliness of the rollers. The solution may leave behind a sticky polymer residue which becomes especially thick at the nips, where the rollers meet. These areas may have enough sticky material to affect the operation of the motors. This can be used to detect over-current, although other methods could be used.

(40) FIG. 21 shows one embodiment of a method to monitor the cleanliness of the rollers. At 70, the rollers would be run 'dry' with no product solution to determine if they are 'sticky.' The motor current or speed would be monitored, and the system microcontroller would be programmed to determine if the rollers are ready to be used for a spray run using product solution at 72. If the rollers are clean and ready to run, the operation of the device is allowed at 74. If they are not ready to be run, the user would be directed to clean the rollers.

(41) Cleaning the rollers may involve replacing the rollers, as in the head cartridge replacement discussed above at 78, the user can insert the roller into a cleaning/docking station to allow the rollers to be cleaned at 79, or the user could insert a cartridge of cleaning solution into the dispenser to clean the rollers with a cleaning solution.

(42) A docking station may be used for other reasons, such as charging, storage, etc., but it can also be used to clean the dispenser. FIGS. 22-27 show embodiments of a docking/cleaning station showing various configuration and locations of the cleaning solution. In FIGS. 22-27, the docking station 80 receives the dispenser casing 12. The dispenser casing has the reservoir 16, the battery and electronics 26 and 28, shown here as the battery 28, and the rollers such as 42. The docking station 80 has station electronics 82, which will typically include a recharging point for the dispenser, either by a connector, contact pads, or an inductive charging system. The docking station will connect to wall power at 88, typically an alternative current source. The docking station will also include a cleaning reservoir 84 of cleaning solution and a waste collection area 86.

(43) FIG. 22 shows a configuration in which the cleaning solution enters the device from the front side of the device. An internal flow channel directs the flow to the rollers where the fluid is dispensed onto the rollers. The fluid may be dispensed similar to the manner in which the fluid is dispensed during spray operation. The docking station, which can communicate the device, either wirelessly or through the electrical contacts, may activate the rollers to clean them. The rollers may spin faster, slower, or the same rate as it would during spray operation based on the cleaning solution being used. In one embodiment, saline solution is used and the rollers spin at a faster rate. The cleaning solution cleans not just the rollers, but is dispersed along the entire nozzle and cleans the inside surface.

(44) In FIG. 23, the cleaning solution is sprayed up into the front side of the rollers by pump 92, or the rollers may spin, and the solution then runs down into the waste collection area 86.

(45) FIG. 24 shows another alternative, where the dispenser 'snaps' into the cleaning station and held in place by a latch or magnet 90. A pump 92 pumps the cleaning solution up to the back side of the rollers. The rollers are activated and spun at very high speed to propel the cleaning solution towards the cleaning station 80. After hitting the back wall of the cleaning station, the excess fluid runs down the system to the waste collection area 86.

(46) FIG. 25 shows an embodiment of the docking station in which the station opens to allow the dispenser to be inserted and then the cleaning solution reservoir closes behind it to allow the cleaning solution to be attached to the back side of the dispenser. In this embodiment the cleaning solution is pressurized to allow the cleaning solution to be sprayed onto the rollers.

(47) FIG. 26 shows an embodiment in which the product fluid reservoir is removed and the device is docked with the cleaning station which includes a feature that mates into the replaceable cartridge hole. Through this mating, it can run cleaning solution through the entire system. The mated feature may be the same size and design as a cartridge or may be significantly larger, only designed to replicate the mating features of the cartridge. The cleaning solution is moved through the system by the pump inside the device, activated by the cleaning station. The cleaning solution is

sprayed or dispersed by the rollers when they are activated and fluid is dispensed onto them. The dispenser has contacts or an inductive charging system in the station electronics **82**.

(48) FIG. **27** shows another embodiment, in which the cleaning solution resides in the docking station and sprays on the front side of the rollers. A magnet or a latch **92**, as shown in FIGS. **24** and **26** is included in the docking station and the device itself. The two parts are designed to mate with each other such that the device is attached to the docking station, but can be removed by the user when cleaning or charging is complete. A tube is included that sprays the solution at a distance close to the rollers themselves. The rollers may be activated so that the material is sprayed or dispersed throughout the device and eventually onto the station itself. The solution then goes into an internal waste collection reservoir connected to the other waste collection area. Any of these docking stations will allow the rollers to be cleaned to allow better operation of the dispenser.

(49) It will be appreciated that variants of the above-disclosed and other features and functions, or alternatives thereof, may be combined into many other different systems or applications. Various presently unforeseen or unanticipated alternatives, modifications, variations, or improvements therein may be subsequently made by those skilled in the art which are also intended to be encompassed by the following claims.

Claims

1. A docking station for a hand-held filament extension atomizer device, comprising: a receiver to receive the device, the device comprising a reservoir of product solution, and a pair of rollers positioned to receive the product solution and diverge to form filaments from the product solution; station electronics; a recharging point for the device arranged to connect with a power source of the device; a power connection to an alternating current power source; a cleaning reservoir of cleaning solution; a pump configured to spray the cleaning solution onto the rollers; and a waste collection area positioned to collect cleaning solution after the cleaning solution has contacted the rollers.
 2. The docking station as claimed in claim 1, wherein the recharging point comprises one of a connector, contact pads or an inductive charging system.
 3. The docking station as claimed in claim 1, wherein the cleaning solution comprises saline.
 4. The docking station as claimed in claim 1, wherein the receiver comprises a latch or magnet to hold the device in place in the docking station.
 5. The docking station as claimed in claim 1, wherein the receiver comprises an internal insertion point for the device and the docking station opens to allow access to the insertion point.
 6. The docking station as claimed in claim 1, wherein the receiver comprises a feature configured to mate with a replaceable cartridge hole on the device.
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